

When buying a digitiser

When buying a digitiser for your application, the digitiser specification should match the application waveform. According to industry experts, following are the top features to consider while buying a new digitiser:

Bandwidth. Any signal consists of a main signal along with its infinite harmonic components. A digitiser has to capture the input wave along with several harmonics in order to get its true shape with least distortions and the most accurate measurements. It is best if the bandwidth of the digitiser is at least two times the highest frequency component of the signal to be measured.

Sampling rate. This is the rate at which the ADC in a digitiser can digitise the incoming signal. Although, according to Nyquist theorem, the sample rate of a digitiser should be just twice the highest frequency component of the input signal, clean representation of a signal requires a sampling rate more than that.

A digitiser with a sampling rate three to four times the highest input frequency is particularly important in time-domain applications such as overshoot, rise time and other pulse measurements. But for applications that require precise timing and amplitude measurements, Sheikh Ahmed of Kingsly Instrumentation & Communication recommends a sampling rate of minimum eight times the highest frequency input.

Resolution and dynamic range. Resolution of an ADC is the number of bits it uses to digitise input samples. If the resolution is n bits, then the input signal can be represented in 2^n number of discrete digital levels or steps.

Dynamic range is the ability of a digitiser to measure a smaller signal in the presence of a larger signal. In other words, it is the ratio of the maximum and minimum signal that can be measured by the digitiser at a single time, and is expressed in decibels. "A higher dynamic range allows simultaneous measurement of signals that contain both large and small voltage components," says Chinmay Misra of

TABLE I
Comparison of Digitisers and DAQ Systems

Feature	Digitiser	DAQ device
Sampling rate	Higher (typically 20MS/s and above)	Lower (typically 10MS/s or less)
Sampling type	Simultaneous sampling	Multiplex sampling mostly; simultaneous sampling rarely
Onboard memory	Larger	Smaller
Timing circuitry	More accurate	Less accurate
Rearm time (time between two triggers)	Faster	Slower
Additional functionalities	Few digital I/Os	Digital I/O channels, analogue I/O channels, counters/timers
Number of channels	Less	More
Vertical range (input voltage range that can be measured)	Wider (up to 50V)	Smaller (about 10V)

TABLE II
Comparison of Digitisers and Digital Oscilloscopes

Feature	Digitiser	Oscilloscope
Application	Large data processing	Troubleshooting
Waveform analysis		Rich set of functions like multiple triggering options, advanced signal acquisition and conditioning modes, waveform analysis; faster
Onboard processing capability	Limited	Better
Integration	Better integration with systems for external processing	
Measurement throughput	High	Low
Flexibility	Less	More
User interface (UI)	Poor UI	Better UI, interactive viewing
Cost of ownership	Lower	High



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NI, "and is important when measuring the overshoot, for example, when testing power supplies and multi-chop packages."

Resolution and dynamic range are related by the following equation:

Dynamic range = maximum signal value : minimum signal value = $2^n : 1$

In normal cases, the dynamic range gets better with higher resolu-

tion. But it is to be noted that the sampling rate decreases with resolution. "It is also to be noted that using an ADC with a higher resolution does not necessarily guarantee higher dynamic range; it depends on effective number of bits (ENOB), which, in turn, depends on the quality of the hardware and the amount of noise it contributes to the measured signal," says Sheikh Ahmed.